

KAMALASAI, Thailand

WMO: 483900

Lat: 16.3319N

Lon: 103.5875E

Elev: 462

StdP: 14.45

Time Zone: 7.00 (E07)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	56.5	59.5	47.8	49.9	63.1	50.6	55.5	64.8	17.8	73.1	14.4	73.1	2.0	N/A	0.342

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	19.1	100.2	79.6	98.1	79.3	96.1	79.2	82.8	91.3	82.2	90.6	81.7	89.9	6.2	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.0	163.8	86.4	80.4	160.5	85.7	79.9	157.7	85.3	47.3	91.7	46.6	90.7	46.0	90.1	88.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.0	12.3	11.2	51.2	104.0	2.9	2.0	49.2	105.4	47.5	106.5	45.9	107.7	43.8	109.1		
(5)			48.4	84.9	2.6	1.7	46.5	86.2	44.9	87.2	43.5	88.1	41.5	89.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	81.5	74.8	77.9	83.5	86.7	86.2	85.1	83.6	82.9	82.5	81.5	78.8	74.8
	DBStd	5.36	4.70	5.51	4.22	3.93	2.85	2.61	3.22	2.15	2.04	2.36	3.36	5.05
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD65	7	3	2	0	0	0	0	0	0	0	0	0	2
	CDD50	11509	770	782	1037	1100	1122	1054	1040	1021	974	978	863	768
	CDD65	6041	308	364	572	650	657	604	575	556	524	513	413	305
	CDH74	65196	2630	3447	6443	8210	8072	7296	6411	5980	5398	5099	3664	2546
	CDH80	28010	966	1517	3245	4451	3991	3279	2525	2085	1765	1888	1387	911
Wind	WSAvg	4.1	4.7	4.7	3.9	3.4	3.8	4.1	4.1	4.0	3.4	3.6	4.4	4.9
Precipitation	PrecAvg	55.2	0.4	0.7	1.7	3.0	7.0	7.9	9.0	10.6	9.8	4.2	0.8	0.1
	PrecMax	67.4	3.2	3.4	4.4	7.0	11.7	13.3	20.4	18.9	16.5	11.7	3.0	0.4
	PrecMin	44.7	0.0	0.0	0.1	0.4	1.8	2.7	3.1	3.0	3.1	0.4	0.0	0.0
	PrecStd	5.7	0.7	1.1	1.3	1.8	2.6	3.0	3.7	3.8	3.5	3.1	0.9	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	91.6	96.3	100.7	103.6	101.8	98.6	99.5	93.5	92.5	92.9	93.0	92.6
		MCWB	74.2	75.6	77.5	81.4	81.1	81.3	80.9	81.3	80.5	80.0	76.3	76.9
	2%	DB	89.8	93.9	98.4	101.0	98.6	95.7	94.6	91.6	90.9	91.2	90.6	89.9
		MCWB	73.0	75.5	77.0	79.9	80.6	80.5	80.3	81.1	80.4	79.1	76.2	75.2
	5%	DB	88.0	92.0	96.2	98.8	96.2	93.7	92.0	90.0	89.5	89.8	89.0	87.9
		MCWB	72.2	74.9	77.4	79.7	80.5	80.5	80.1	80.7	80.1	78.2	75.9	74.2
	10%	DB	85.9	89.2	93.8	96.2	93.7	91.9	89.6	88.4	87.9	88.4	87.4	85.4
		MCWB	71.1	73.8	77.5	79.5	80.4	80.5	79.9	80.3	80.0	77.7	75.3	72.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	77.9	79.1	81.7	84.4	83.7	83.4	82.7	83.1	82.7	81.6	80.0	78.2
		MCDB	86.6	89.0	91.2	96.2	92.5	91.6	90.5	89.4	88.7	89.1	87.7	89.7
	2%	WB	76.3	77.9	80.4	82.4	82.6	82.5	81.8	82.2	81.8	80.6	79.0	76.9
		MCDB	84.4	88.0	91.0	94.2	92.2	90.7	89.7	88.5	87.5	87.6	86.4	86.8
	5%	WB	74.7	76.9	79.5	81.4	81.8	81.8	81.0	81.5	81.3	79.9	78.0	75.4
		MCDB	83.6	87.2	90.7	93.4	91.4	89.9	88.5	87.6	86.7	86.4	84.9	84.5
	10%	WB	73.1	75.9	78.6	80.5	81.1	81.2	80.5	80.8	80.7	79.1	76.9	74.2
		MCDB	82.3	85.6	90.1	92.5	90.5	89.1	87.7	86.7	85.9	85.2	83.4	82.8
Mean Daily Temperature Range	5% DB	MDBR	21.9	21.6	20.2	19.1	15.9	13.2	11.4	10.9	11.0	14.0	17.2	20.2
		MCDBR	22.6	22.7	23.0	21.7	18.7	15.8	15.0	13.2	13.1	15.2	18.3	21.1
		MCWBR	8.5	7.5	6.8	6.3	5.3	4.7	4.7	5.1	5.2	5.6	6.5	8.1
	5% WB	MCDBR	19.1	19.9	20.6	20.3	16.7	14.0	12.5	12.3	12.0	13.5	15.4	19.6
MCWBR		8.4	7.1	7.2	6.6	6.0	5.2	4.9	5.2	5.3	5.3	6.1	8.3	
Clear-Sky Solar Irradiance	taub	0.475	0.592	0.664	0.624	0.516	0.471	0.459	0.465	0.508	0.527	0.470	0.466	
	taud	2.090	1.797	1.679	1.808	2.131	2.279	2.312	2.284	2.150	2.062	2.196	2.142	
	Ebn at Noon	257	232	220	230	252	262	265	266	255	246	257	255	
	Edh at Noon	49	69	80	70	50	43	42	43	49	52	44	46	
All-Sky Solar Radiation	RadAvg	1554	1643	1677	1811	1752	1634	1524	1490	1463	1596	1563	1501	
	RadStd	89	82	77	81	95	84	113	96	108	107	54	38	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (21 neighbors)	+0.50	N/A	N/A	N/A	+0.23	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page